AIM: To create a Hospital Management System using Kotlin

Program:

Main.kt

```kotlin
import java.util.Scanner

data class Patient(  3 Usages
    val id: Int,
    val name: String,
    val disease: String,
    var admitted: Boolean = true
)

class Hospital {  1 Usage
    private val patients = mutableListOf<Patient>()  5 Usages

    fun addPatient(patient: Patient) {  1 Usage
        patients.add(patient)
        println("Patient added successfully!")
    }

    fun displayPatients() {  1 Usage
        if (patients.isEmpty()) {
            println("No patients available.")
            return
        }
        for (p in patients) {
            println("ID: ${p.id}, Name: ${p.name}, Disease: ${p.disease}, Admitted: ${p.admitted}")
        }
    }

    fun searchPatient(name: String) {  1 Usage
        val found = patients.filter { it.name.equals( other = name, ignoreCase = true) }
        if (found.isEmpty()) {
            println("Patient not found.")
```

```
    fun searchPatient(name: String) {  1 Usage
        } else {
            for (p in found) {
                println("Found -> ID: ${p.id}, Disease: ${p.disease}, Admitted: ${p.admitted}")
            }
        }
    }

    fun dischargePatient(id: Int) {  1 Usage
        val patient = patients.find { it.id == id }
        if (patient != null && patient.admitted) {
            patient.admitted = false
            println("Patient discharged successfully!")
        } else {
            println("Invalid ID or already discharged.")
        }
    }
}

fun main() {
    val scanner = Scanner( source = System.`in`)
    val hospital = Hospital()
    var choice: Int

    do {
        println("\n--- Hospital Management System ---")
        println("1. Add Patient")
        println("2. Display Patients")
        println("3. Search Patient")
        println("4. Discharge Patient")
        println("5. Exit")
```

```
        print("Enter your choice: ")

        choice = scanner.nextInt()

        when (choice) {
            1 -> {
                print("Enter Patient ID: ")
                val id = scanner.nextInt()
                scanner.nextLine()

                print("Enter Name: ")
                val name = scanner.nextLine()

                print("Enter Disease: ")
                val disease = scanner.nextLine()

                hospital.addPatient(Patient(id, name, disease))
            }

            2 -> hospital.displayPatients()

            3 -> {
                scanner.nextLine()
                print("Enter name to search: ")
                val name = scanner.nextLine()
                hospital.searchPatient(name)
            }

```

```
            4 -> {
                print("Enter Patient ID to discharge: ")
                val id = scanner.nextInt()
                hospital.dischargePatient(id)
            }

            5 -> println("Exiting... Thank you!")

            else -> println("Invalid choice!")
        }

    } while (choice != 5)
}
```

OP:

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\User

--- Hospital Management System ---
1. Add Patient
2. Display Patients
3. Search Patient
4. Discharge Patient
5. Exit
Enter your choice: 1
Enter Patient ID: 23
Enter Name: payal
Enter Disease: cancer
Patient added successfully!

--- Hospital Management System ---
1. Add Patient
2. Display Patients
3. Search Patient
4. Discharge Patient
5. Exit
Enter your choice: 2
ID: 23, Name: payal, Disease: cancer, Admitted: true
```

```
--- Hospital Management System ---
1. Add Patient
2. Display Patients
3. Search Patient
4. Discharge Patient
5. Exit
Enter your choice: 3
Enter name to search: payal
Found -> ID: 23, Disease: cancer, Admitted: true

--- Hospital Management System ---
1. Add Patient
2. Display Patients
3. Search Patient
4. Discharge Patient
5. Exit
Enter your choice: 4
Enter Patient ID to discharge: 23
Patient discharged successfully!
```

```
--- Hospital Management System ---
1. Add Patient
2. Display Patients
3. Search Patient
4. Discharge Patient
5. Exit
Enter your choice: 5
Exiting... Thank you!

Process finished with exit code 0
```